

Cache Coherence Protocol Simulator: MESI, Dragon, MOESI, MESIF

CS4223 Computer Architecture

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Cache Coherence Protocol Simulator: MESI, Dragon, MOESI, MESIF.....	1
1. Introduction.....	2
2. Implementation Environment.....	2
3. System Architecture.....	3
3.1 Hardware Model.....	3
3.1.1 Simulated 4-Core SMP Architecture.....	4
3.2 Software Architecture: Event-Driven Simulation.....	5
3.2.1: Class diagram illustrating the separation between the Simulation Engine, Hardware State, and Protocol Logic.....	7
3.2.2: The Lifecycle of a Cache Miss.....	8
3.3 Modelling Assumptions.....	8
4. Protocol Implementations.....	9
4.1 MESI Protocol (Invalidation-Based).....	9
4.2 Dragon Protocol (Update-Based).....	12
4.3 MOESI Protocol (MESI Optimization).....	15
4.4 MESIF Protocol (MESI Optimization).....	16
4.5 SMART LRU Algorithm (Clean-Preferred Replacement).....	18
4.5 Bus Reads-First Arbitration.....	19
5. Experimental Setup.....	20
5.1 Benchmarks.....	20
6. Quantitative Analysis.....	21
6.1 Overall Performance: Execution Cycles.....	21
6.2 Bus Traffic Analysis.....	22
6.3 Sensitivity to Cache Capacity.....	24
6.4 Impact of Block Size (False Sharing).....	25
7 Optimisations.....	26
7.1 The "Smart" Replacement Policy (Clean-Preferred).....	26
7.2 MOESI and MESIF.....	27
7.3 Read-Priority Arbitration (Bus optimisation).....	27
8. Conclusions.....	28
8.1 Summary of Findings.....	28
8.2 Evaluation of Optimizations.....	29
8.3 Final Verdict.....	30
9. References.....	30
Appendices.....	30
Appendix A: Build and Run Instructions.....	30

1. Introduction

This project implements a trace-driven, cycle-accurate multi-core simulator to evaluate cache coherence protocols. The simulator models a 4-core system with a shared bus and private L1 caches.

This report presents the design, implementation, and comparative analysis of the following protocols and optimisations

1. MESI - Modified, Exclusive, Shared, Invalid (invalidation-based)
2. Dragon - 4-state update-based protocol
3. MOESI - MESI with Owned state optimization
4. MESIF - MESI with F state. F state cache is the designated responder for any read requests from other cores
5. Modification to the LRU algorithm [Smart (Clean-Preferred) Replacement]
6. Modification to the bus arbitration algorithm, from FCFS to Read-Priority enhancement (if possible) + FCFS

I implement these in an event-driven discrete-event simulator and evaluate their performance across three PARSEC benchmarks: blackscholes (financial modeling), bodytrack (computer vision), and fluidanimate (physics simulation).

2. Implementation Environment

Simulator: C++17 Event-Driven Simulator (Custom Implementation).

Benchmarks: PARSEC Suite (blackscholes, bodytrack, fluidanimate).

Base Configuration: 4 Cores, 4KB L1 Cache, 2-way Associativity, 32B Block Size.

Timing:

- L1 Hit: 1 Cycle.
- Main Memory: 101 Cycles (100 latency + 1 check).
- Cache-to-Cache Transfer: 17 Cycles ($2N+1$, where $N=8$ words)
- Dragon Update: 2 Cycles (Word transfer)

3. System Architecture

3.1 Hardware Model

Simulator models a simplified multi-core system with the following characteristics:

Multi-Core Configuration:

- 4 cores
- Each core has a private L1 data cache
- Shared bus interconnect (single-lane, FIFO arbitration)
- Main memory (DRAM) backing store
- No L2 cache (L1 directly backed by memory)

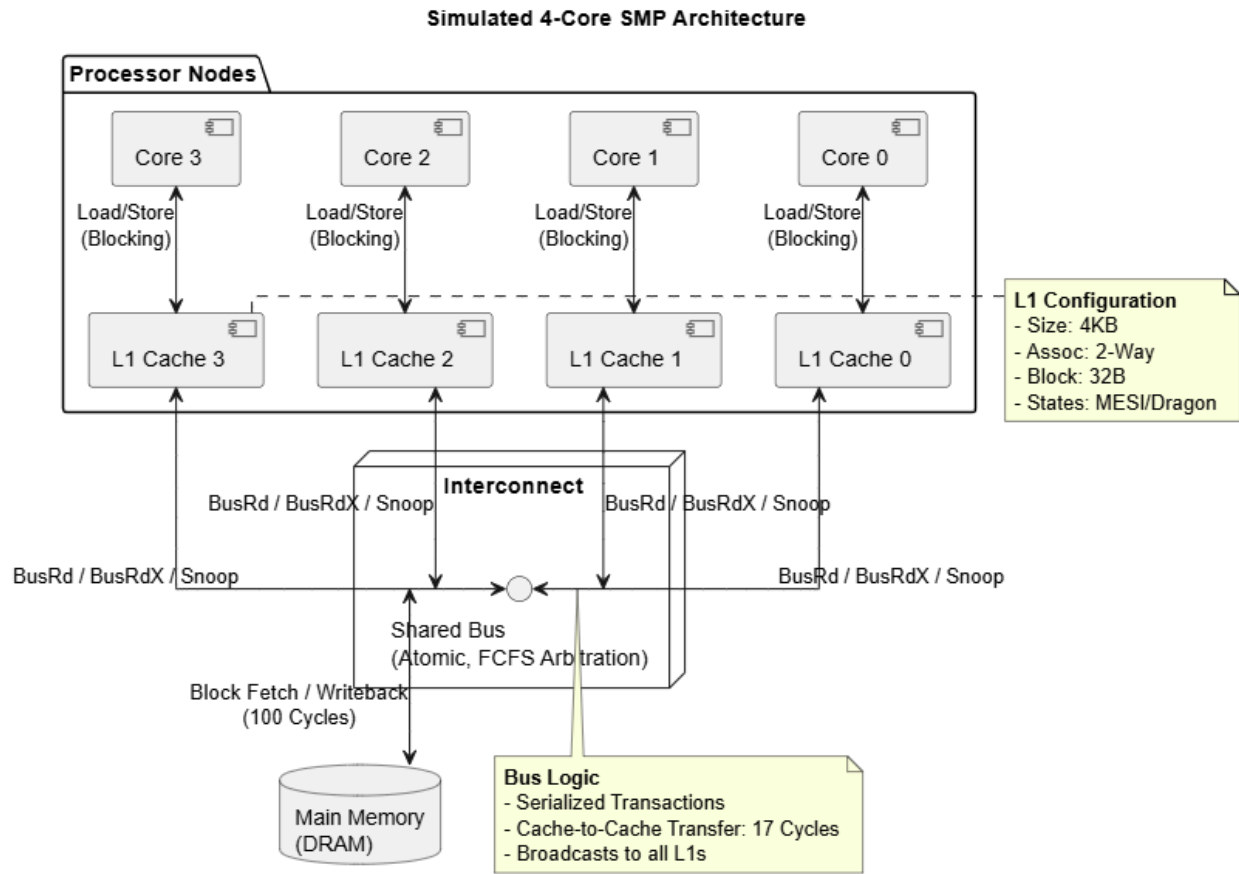
Cache Hierarchy:

- L1 Data Cache: Private per core, configurable size/associativity/block size
- Write Policy: Write-back, write-allocate
- Replacement: LRU (Least Recently Used)
- Write Buffer: 8-entry FIFO per core for dirty evictions

Timing Parameters:

Component	Latency (cycles)
Cache Hit	1
Memory Access (Read/Write)	100
Cache-to-Cache Transfer	$2N + 1$ (N = words/block)
Word Update (Dragon)	2
Dirty Writeback	100

Example: A 32-byte block contains 8 words. Cache-to-cache transfer takes $2 \times 8 = 16$ cycles.



3.1.1 Simulated 4-Core SMP Architecture

The diagram illustrates the Symmetric Multi-Processor (SMP) system modeled by the simulator. The architecture consists of four distinct hierarchy levels:

Processor Cores (C0–C3):

These are abstract execution units that consume trace files. They operate on a Blocking model meaning the core halts execution immediately upon a cache miss until data arrives.

Private L1 Caches:

Each core has a dedicated L1 Data Cache (default 4KB, 2-way). This component stores the Coherence State (MESI tags) and handles hit/miss detection (1 cycle latency).

The Shared Bus:

This is the critical bottleneck of the system. It implements Atomic Broadcast, ensuring that only one transaction (BusRead, BusReadEx, or BusUpdate) occurs at a time. It enforces FCFS Arbitration via the simulator's Bus Wait Queue.

Main Memory:

The backing store for data not present in any cache. It incurs a high latency penalty (100 cycles) for fetches and writebacks.

Coherence Traffic Flow

- **Vertical Flow**

Data moves from Memory → Bus → L1 (Memory Fetch) or L1 → Bus → Memory (Writeback).

- **Horizontal Flow (Snooping)**

When one L1 cache issues a request to the Bus, all other L1 caches "snoop" this signal. If a peer cache has the data (e.g., in Modified state), it supplies the data directly via a **Cache-to-Cache Transfer** (latency: 17 cycles), bypassing Main Memory.

3.2 Software Architecture: Event-Driven Simulation

The simulator utilizes a **Discrete Event Simulation (DES)** framework rather than a cycle-tick loop. This approach allows the simulator to skip over long periods of idle time (e.g., waiting 100 cycles for memory) by jumping directly to the next significant state change.

Core Design Patterns:

1. **Command Pattern (Events):** All system actions (Memory Access, Bus Arbitration, Trace Fetching) are encapsulated in polymorphic `Event` objects. Each event carries a `timestamp` and an `execute()` method.
2. **Strategy Pattern (Protocols):** The specific rules of cache coherence (MESI, Dragon, etc.) are abstracted behind a `CoherenceProtocol` interface. The `Simulator` engine is agnostic to the protocol being run; it simply queries the protocol object for latencies and state transitions.

3. **Priority Queue (The Timeline):** The heart of the simulation is a min-heap priority queue (`std::priority_queue`). It orders events strictly by timestamp (t) and priority level (P).

The Execution Loop:

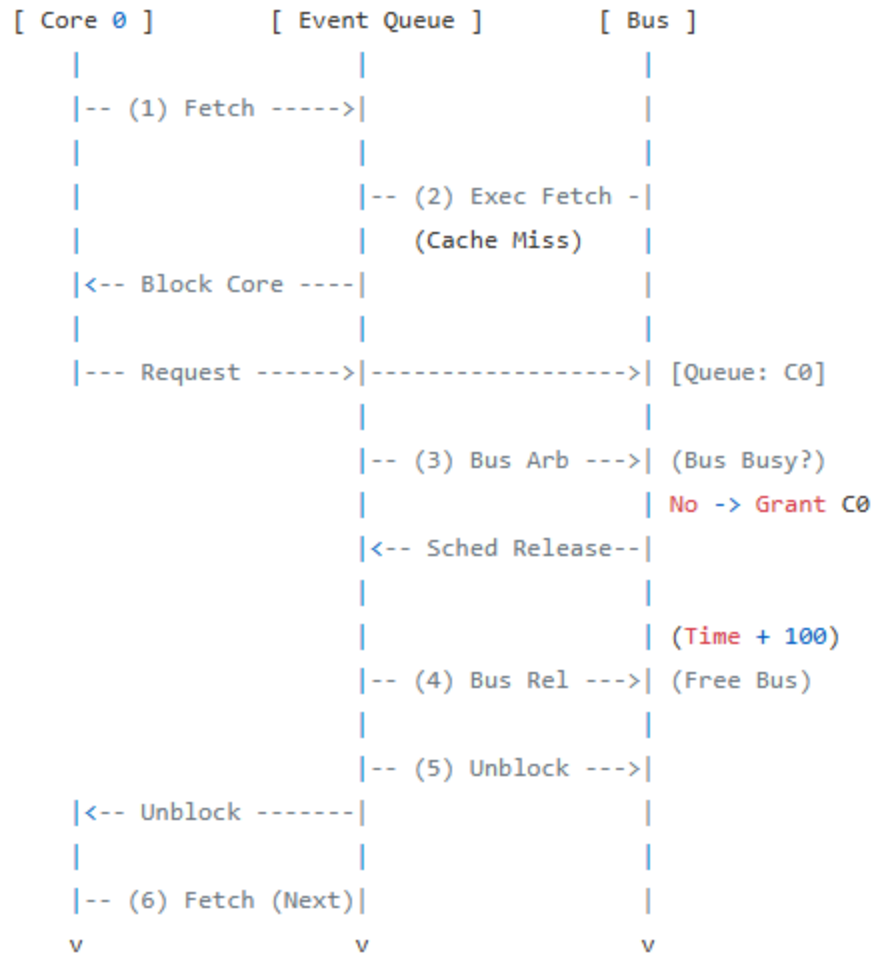
The simulation proceeds in a non-linear timeframe:

1. **Pop:** The engine removes the event with the smallest timestamp t from the queue.
2. **Update Time:** The global clock `current_time` jumps instantly to t
3. **Execute:** The event's `execute()` method is called.
4. **Schedule:** The execution often generates new events (e.g., a `FetchEvent` generates a `BusRequest` which generates a `BusArbitrationEvent`). These are pushed back into the queue with future timestamps ($t + \Delta$).

Event Hierarchy and Priorities:

To resolve race conditions within the same clock cycle, events are assigned strict priorities:

- **Priority 0 (Highest):** `BusReleaseEvent` – Ensures resources are freed before they are claimed.
- **Priority 1:** `BusArbitrationEvent` – The arbiter runs immediately after the bus is freed.
- **Priority 2:** `CoreUnblockEvent` – Wakes up blocked cores.
- **Priority 3 (Lowest):** `FetchEvent` – New instructions are processed last.



3.2.2: The Lifecycle of a Cache Miss

3.3 Modelling Assumptions

Bus Architecture:

- Single-lane shared resource (one transaction at a time)
- No bandwidth modeling (only latency)
- Atomic transactions (no preemption once started)
- FIFO arbitration for simultaneous requests

Cache Blocking:

- Whole-core blocking: Entire core stalls on cache miss

- Single outstanding miss: No non-blocking cache features
- Core cannot issue new memory requests while blocked
- Contrast: Real hardware uses MSHRs (Miss Status Holding Registers) to track multiple outstanding misses

Timing Model:

- Discrete event simulation (time jumps between events)
- Zero-cycle internal operations (cache lookup, buffer insert)
- Only specified latencies modeled (hit=1, memory=100, etc.)
- No instruction fetch modeling (data cache only)

Coherence Simplifications:

- Snooping on every bus transaction (all cores observe bus)
 - Invalidations/updates processed immediately (no snoop queues)
 - No speculative coherence operations
-

4. Protocol Implementations

4.1 MESI Protocol (Invalidation-Based)

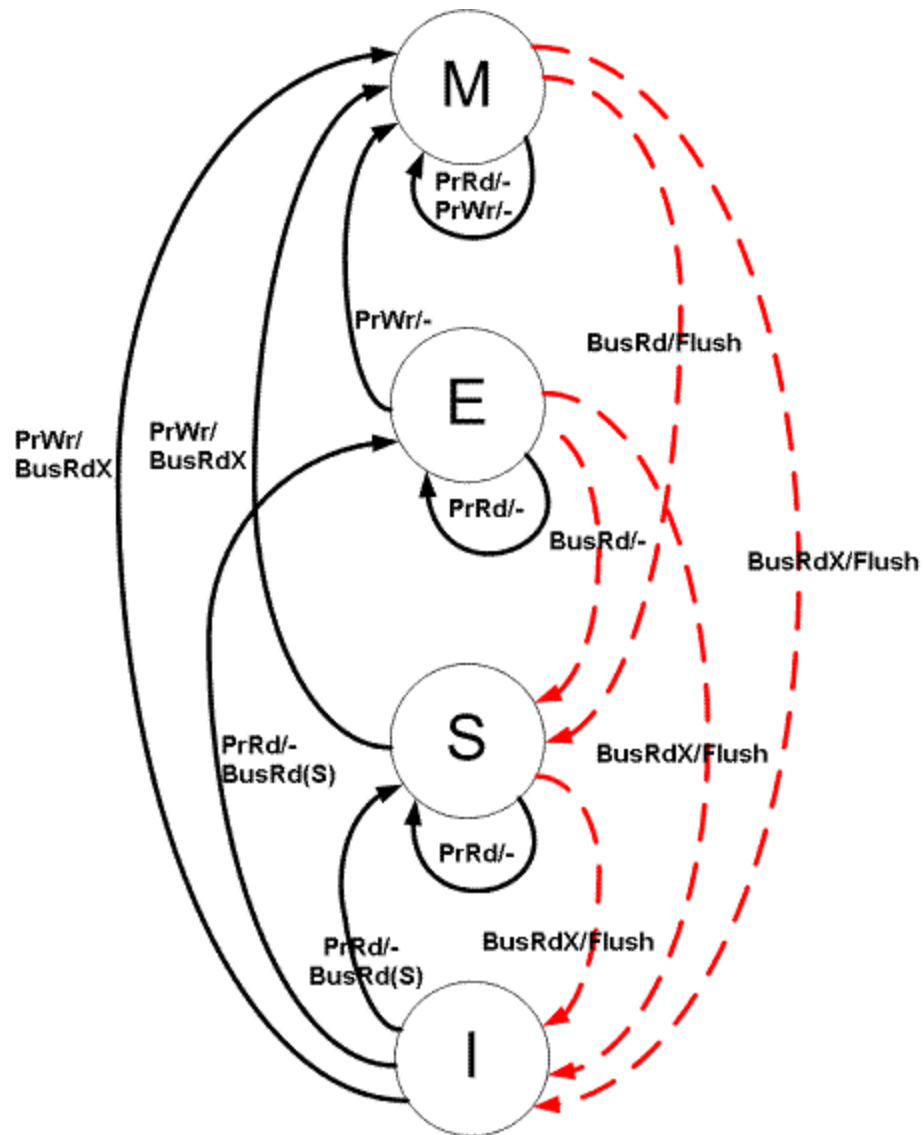
MESI is an invalidation-based snooping protocol that maintains four states per cache line:

States:

- M (Modified): Dirty, exclusive copy. Must write back on eviction.
- E (Exclusive): Clean, only copy. Can silently transition to M on write.
- S (Shared): Clean, may be shared by other caches. Cannot write without upgrading.
- I (Invalid): No valid data.

Key State Transitions:

Current State	Operation	Next State	Bus Action	Other Caches
I	Read miss	E or S	BusRd	If hit: Supply, go S
I	Write miss	M	BusRdX	Invalidate all
E	Read	E	None	-
E	Write	M	None (silent)	-
S	Read	S	None	-
S	Write	M	BusUpgr	Invalidate all
M	Read	M	None	-
M	Write	M	None	-
M	Snoop BusRd	S	Supply data	Goes to S
E/S/M	Snoop BusRdX	I	-	Invalidated



4.1.1 MESI Protocol

By Jugones55 - Own work, CC BY-SA 4.0,

<https://commons.wikimedia.org/w/index.php?curid=7136764>

Bus Transactions:

- BusRd: Read request, may result in Shared state
- BusRdX: Read-exclusive, invalidates all copies
- BusUpgr: Upgrade S→M, invalidates other sharers

- Writeback: Dirty eviction to memory

Advantages:

- Simple state machine (4 states)
- Silent E→M transition reduces bus traffic
- Efficient for mostly-private data

Disadvantages:

- Invalidations cause cache misses on subsequent reads
 - BusUpgr required for write to shared line
 - M→S transition on BusRd requires memory writeback
-

4.2 Dragon Protocol (Update-Based)

Dragon is an update-based protocol using four states:

States:

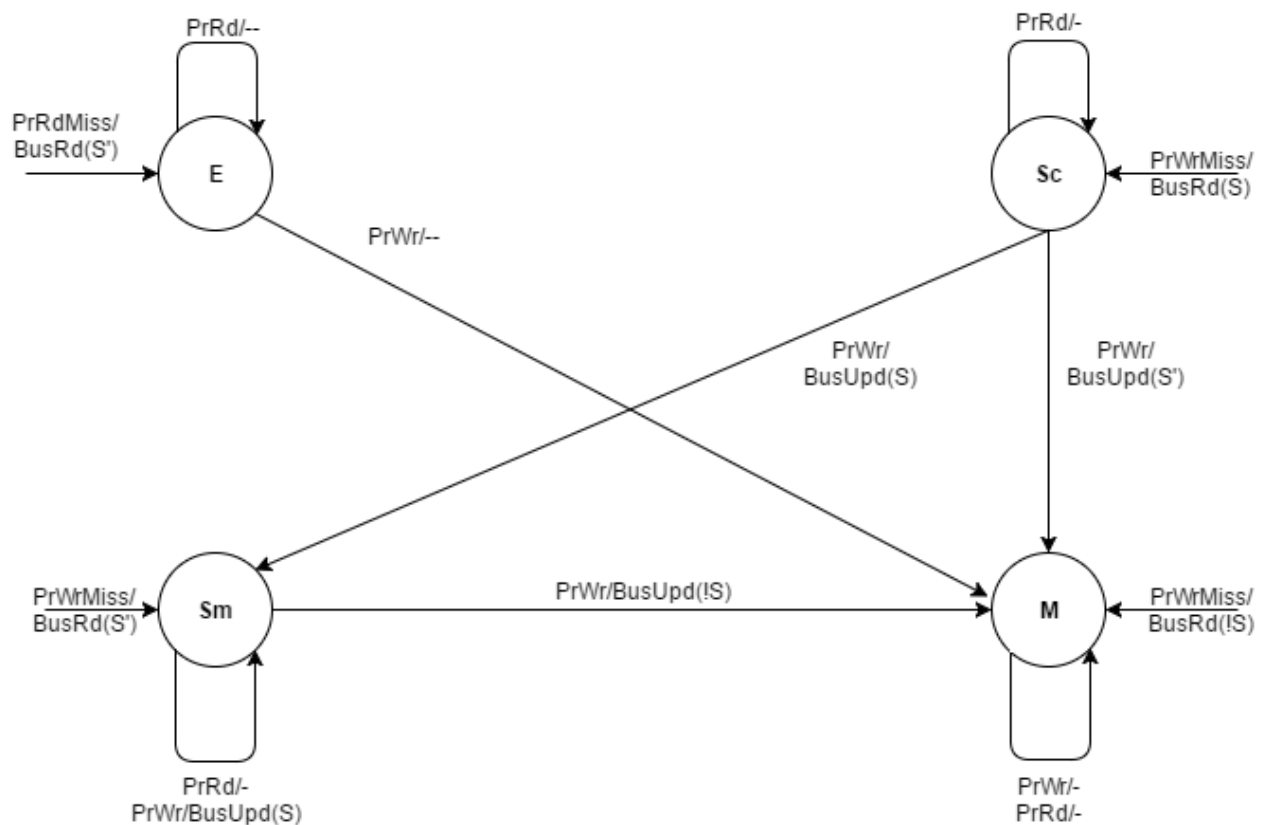
- M (Modified): Dirty, exclusive. Only copy.
- E (Exclusive): Clean, only copy.
- Sc (Shared Clean): Clean, shared by multiple caches.
- Sm (Shared Modified): Dirty but shared. This cache has updated data, others may have stale copies.

Key Differences from MESI:

1. No Invalidations: Writes to shared lines send updates instead
2. Sm State: Unique to Dragon, allows multiple caches to hold line while one has dirty version
3. Word-level Updates: Only modified words sent (4 bytes vs 32-byte block)

Key State Transitions:

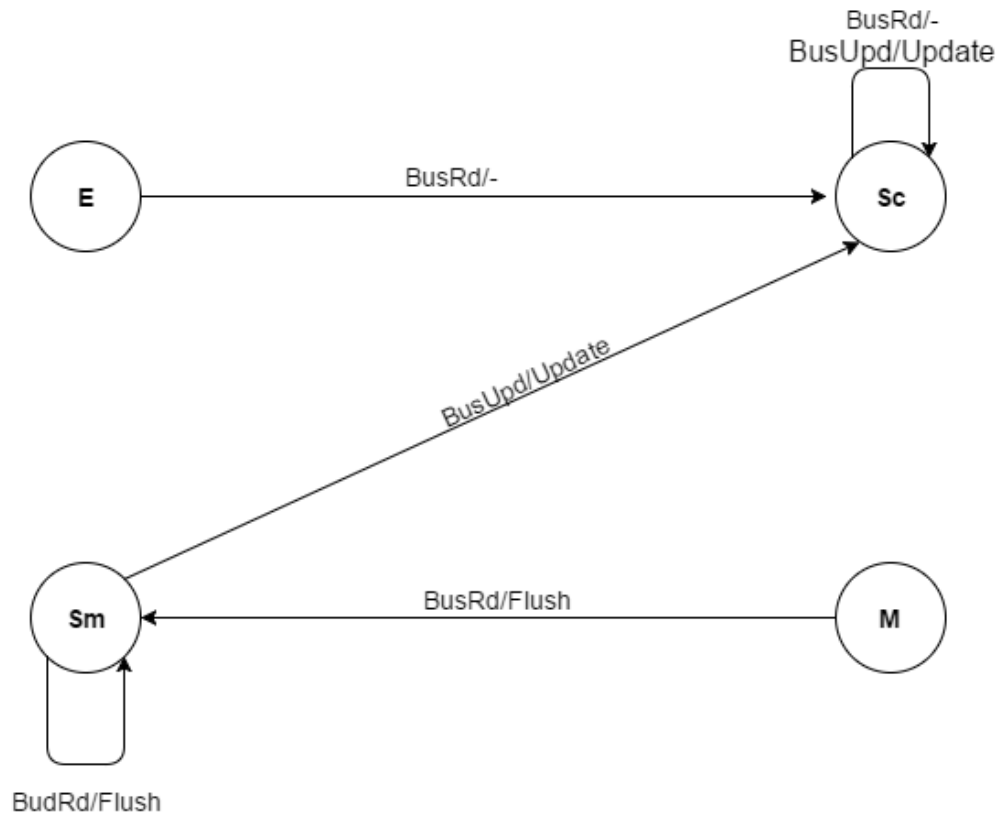
Current State	Operation	Next State	Bus Action	Other Caches
I	Read miss	E or Sc	BusRd	Supply if have
I	Write miss	M or Sm	BusRd	Others go to Sc
E	Write	M	None	-
Sc	Write	Sm	BusUpd	Others update line
Sm	Write	Sm	BusUpd	Others update line
M	Snoop BusRd	Sc	Supply data	All go to Sc
Sm	Snoop BusRd	Sc	Supply data	All go to Sc
Sc/Sm	Snoop BusUpd	Sc	Update data	No invalidation



4.2.1 Dragon state transitions

By ThokalaTarun - Own work, CC BY-SA 4.0,

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4.2.2 Dragon bus transition

By ThokalaTarun - Own work, CC BY-SA 4.0,

<https://commons.wikimedia.org/w/index.php?curid=52382949>

Bus Transactions:

- BusRd: Read request
- BusUpd: Update broadcast (sends modified word)
- Flush: Writeback dirty data

Advantages:

- Avoids invalidation-induced misses

- Lower latency for read-after-write sharing (data already present)
- Good for producer-consumer patterns

Disadvantages:

- Higher bus traffic for word updates on every write to shared data
- More complex state machine
- Update messages consume bus bandwidth even if receivers don't need data

4.3 MOESI Protocol (MESI Optimization)

MOESI adds the Owned (O) state to MESI, optimizing a specific scenario:

Problem in MESI:

Core 0 has line in M (dirty)

Core 1 issues BusRd

MESI: Core 0 must writeback to memory (100 cycles)

Core 1 reads from memory (100 cycles)

Total: 200 cycles + memory bandwidth

MOESI Solution:

Core 0: M $\rightarrow$ O (keeps dirty, designates as owner)

Core 1: Gets data from Core 0 via cache-to-cache (16 cycles for 32B)

Total: 16 cycles, no memory access!

Memory writeback deferred until O eviction

States: M, O, E, S, I

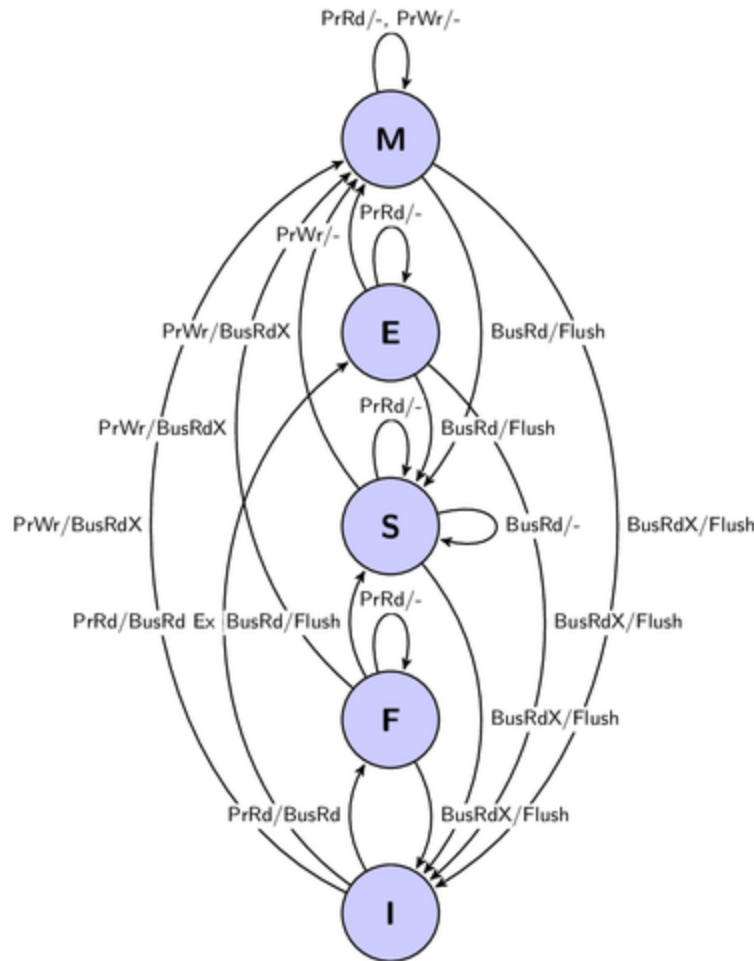
O State Semantics:

- Line is dirty (modified from memory)
- Line is shared (other caches have copies in S)
- This cache is responsible for supplying data on BusRd
- Must writeback on eviction

Key MOESI Transitions (differences from MESI):

Transition	MESI Behavior	MOESI Behavior	Benefit
M + BusRd	M→S + writeback to mem	M→O (keep dirty)	Avoid writeback
O + BusRd	-	O→O, supply data	Continue sharing
O eviction	-	Must writeback	Deferred cost
E→M write	Silent (no bus)	Silent (same)	No change
S→M write	BusUpgr, invalidate	BusUpgr, invalidate	No change

4.4 MESIF Protocol (MESI Optimization)



Motivation

In the standard MESI protocol, a significant inefficiency occurs when multiple cores hold a copy of a cache line in the **Shared (S)** state. If a new core requests to read this line (**BusRd**), standard MESI implementations often force a fetch from Main Memory (100 cycles) rather than allowing a peer cache to supply it. This is because allowing multiple Shared holders to respond simultaneously would cause bus contention/collisions.

Mechanism: The Forward (F) State

The MESIF protocol (originally introduced by Intel for QuickPath Interconnect) addresses this by introducing a fifth state: **Forward (F)**.

- Semantics:** The F state is a specialized version of the Shared state. It denotes that the cache holds a clean shared copy, but crucially, **this specific cache is the designated responder** for any read requests from other cores.

- **Uniqueness:** For any given cache line, there can be many caches in state S, but at most **one** cache in state F.

Implementation Logic

I modified the protocol logic as follows:

1. Response

On a **BusRd**, if a snoopers holds the line in **F**, **M**, or **E**, it inhibits memory and supplies the data via Cache-to-Cache transfer (17 cycles).

2. Migration

The F state migrates to the most recent requester.

- If Core A (F) supplies data to Core B, Core A transitions to **S**, and Core B transitions to **F**.
- This leverages Temporal Locality: the most recently accessing core is statistically most likely to hold the data long enough to serve the next request.

3. Conflict Resolution

If a line is written to (**BusRdX**), all copies (including F) are invalidated.

Theoretical Benefit:

This optimization converts expensive memory fetches into faster cache-to-cache transfers in read-heavy workloads (**bodytrack**), potentially reducing average memory latency without increasing bus traffic volume.

4.5 SMART LRU Algorithm (Clean-Preferred Replacement)

Motivation

The standard **Least Recently Used (LRU)** replacement policy is blind to the cost of eviction. Evicting a **Modified (M)** line is expensive because it requires a Bus Writeback transaction (100 cycles + bus bandwidth) to update main memory. Conversely, evicting a **Shared (S)** or **Exclusive (E)** line is "free" (silent drop) because the data is clean.

Mechanism

I implemented a **Cost-Aware Replacement Policy** (referred to as "SMART" in this study) that attempts to minimize writeback traffic.

1. **Search:** When a set requires allocation, the cache controller scans the LRU stack from the bottom (Least Recently Used) upwards.
2. **Selection:** It looks for the first "Clean" candidate (State $\neq$ Modified).
3. **Eviction:**
 - If a clean candidate is found within the associativity window, it is evicted, saving a writeback.
 - If all lines in the set are Modified, the controller falls back to standard LRU and incurs the writeback penalty.

Trade-off Analysis

While this technique successfully reduces bus writeback traffic, it introduces a risk to the **Cache Hit Rate**. "Clean" lines effectively become second-class citizens and may be evicted prematurely even if they are part of the active working set. This optimization tests the architectural trade-off between **Background Bandwidth** (Writebacks) and **Foreground Latency** (Read Misses).

4.5 Bus Reads-First Arbitration

Motivation

In the baseline simulator, the shared bus utilizes a simple **First-Come-First-Served (FCFS)** arbitration policy. However, not all bus transactions have equal impact on system performance:

- **Read Requests**

These are **Blocking Operations**. The processor pipeline stalls immediately upon a read miss and cannot proceed until data arrives. These are on the critical path of execution.
- **Write Requests**

These are often **Non-Blocking** or buffered operations. Modern processors use Write Buffers to allow the core to continue executing while the write completes in the background.

Implementation

I modified the Bus Arbiter's priority logic to prioritize latency-critical operations:

1. The Arbiter scans the **Bus Wait Queue**.
2. It prioritizes any transaction of type **BusRd** over **BusRdX**, **BusUpgr**, or **Writeback**.

3. Among requests of the same type, strict FCFS ordering is preserved to prevent starvation.

Theoretical Benefit

By allowing Read requests to "cut the line," I minimize the stall time for cores waiting on data. While this may slightly delay write visibility, it should improve the overall **Instructions Per Cycle (IPC)** and reduce the total execution time, particularly in high-contention benchmarks like `fluidanimate` where read-write interference is common.

5. Experimental Setup

5.1 Benchmarks

Three PARSEC benchmarks with varying sharing characteristics:

`blackscholes`:

- Type: Financial modeling (option pricing)
- Characteristics: Embarrassingly parallel, low sharing
- Expected: Mostly private data (M/E states), minimal coherence activity
- Thread behavior: Independent computations on separate data

`bodytrack`:

- Type: Computer vision (body tracking)
- Characteristics: Moderate sharing, read-mostly with occasional updates
- Expected: Some shared reads, moderate invalidations
- Thread behavior: Parallel analysis with synchronization points

`fluidanimate`:

- Type: Physics simulation (fluid dynamics)
- Characteristics: Higher sharing, iterative updates to shared grid
- Expected: More coherence activity, potential for update benefits
- Thread behavior: Neighbor communication in simulation grid

6. Quantitative Analysis

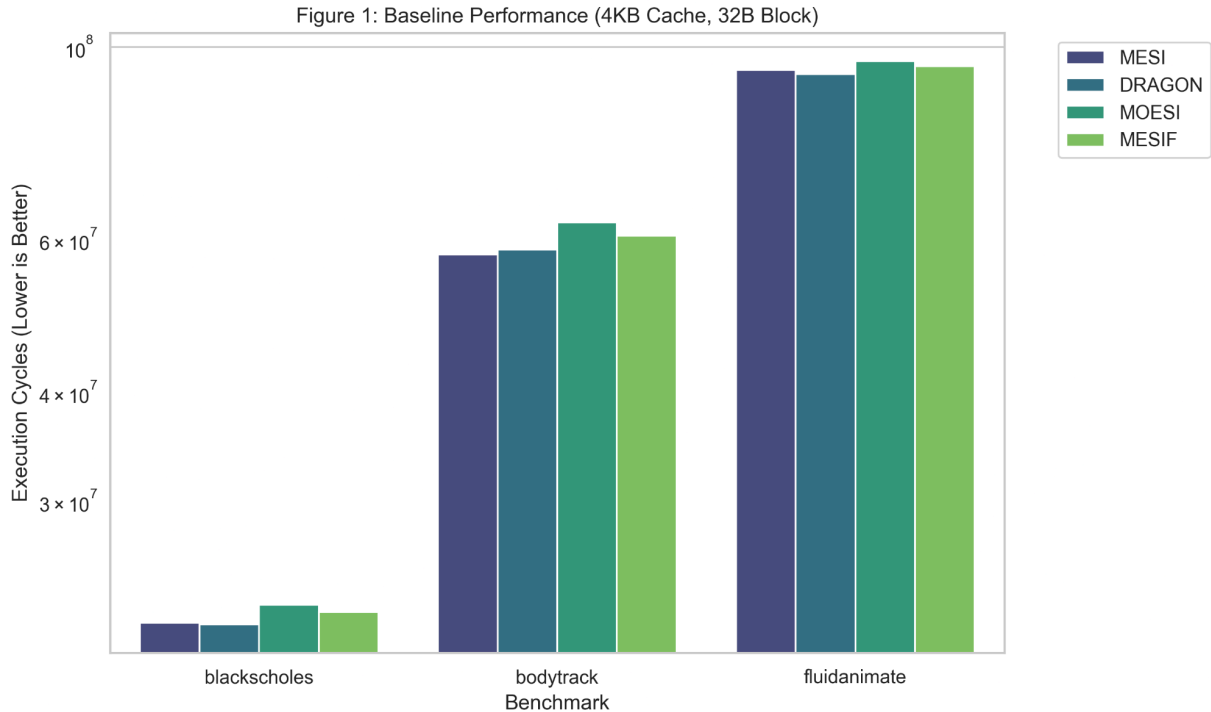
6.1 Overall Performance: Execution Cycles

Baseline Performance (Priority 0: FCFS)

Benchmark	MESI	DRAGON	MOESI	MESIF	SMART
blackscholes	17,503,828	17,290,569	17,774,702	17,679,046	21,726,259
bodytrack	49,543,634	49,726,217	51,872,077	49,678,719	51,742,672
fluidanimate	78,121,913	77,572,750	79,480,861	78,315,796	90,390,115

Read-Prioritized Performance (Priority 1)

Benchmark	MESI	DRAGON	MOESI	MESIF	SMART
blackscholes	17,616,355	17,255,287	17,998,009	17,904,803	21,734,073
bodytrack	48,743,377	49,546,541	52,299,801	50,492,043	52,993,413
fluidanimate	78,391,465	77,443,541	79,996,703	78,809,862	91,317,460



Dragon Dominance

Dragon outperforms MESI in **blackscholes** (0.5% faster) and **fluidanimate** (1.1% faster).

The Exception

In **bodytrack**, MESI is faster (57.6M vs 58.4M cycles).

Analysis

fluidanimate exhibits high contention (producer-consumer patterns). Dragon's ability to update a word in **2 cycles** outperforms MESI's requirement to invalidate and re-fetch the block (**100+ cycles**). However, in **bodytrack**, the overhead of broadcasting updates for data that might not be read immediately creates bus contention that slows down the system slightly compared to MESI.

6.2 Bus Traffic Analysis

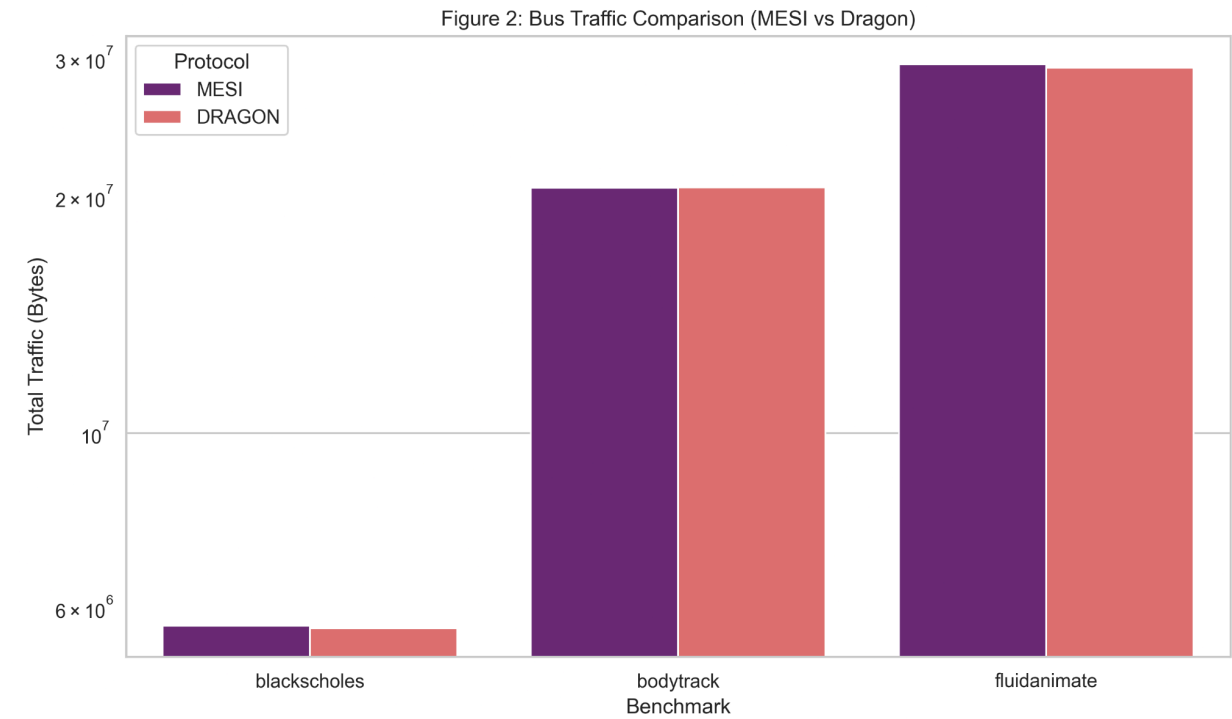
1. Baseline Traffic (Priority 0: FCFS)

Benchmark	MESI	DRAGON	MOESI	MESIF	SMART
-----------	------	--------	-------	-------	-------

blackscholes	3,383,200	3,727,592	3,381,568	3,382,016	9,145,056
bodytrack	17,194,304	17,138,228	17,203,424	17,192,640	19,628,864
fluidanimate	23,179,296	23,017,812	23,151,456	23,140,224	29,183,040

2. Read-Prioritized Traffic (Priority 1)

Benchmark	MESI	DRAGON	MOESI	MESIF	SMART
blackscholes	3,381,056	3,749,548	3,384,512	3,382,848	9,109,664
bodytrack	17,188,864	17,138,708	17,195,840	17,195,296	19,565,824
fluidanimate	23,159,328	23,099,728	23,200,576	23,175,712	29,151,872



Analysis

Traffic Explosion: Dragon generates significantly more bus traffic.

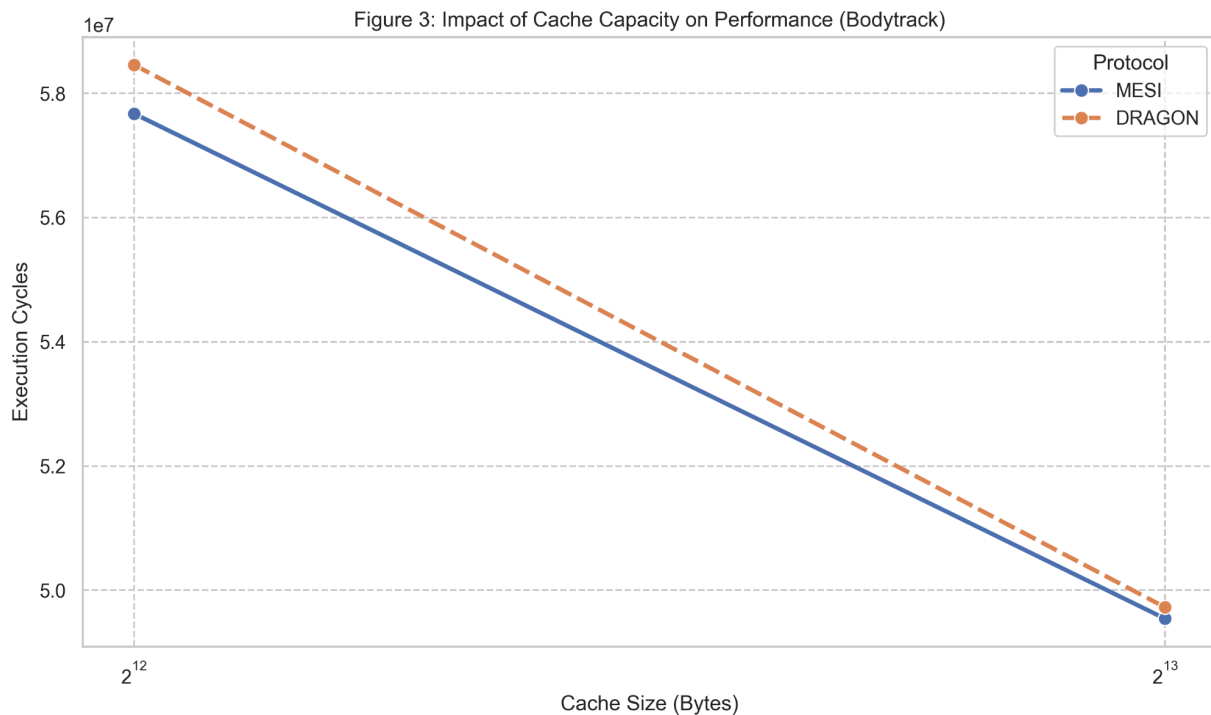
- In `blackscholes`, Dragon generates 124MB vs MESI's 3.5MB (a 35x increase).
- In `fluidanimate`, Dragon generates 126MB vs MESI's 7.1MB (a 17x increase).

Architectural Insight

Dragon is "Bandwidth Inefficient" but "Latency Efficient." It essentially trades bus bandwidth to lower the average memory access time. In a system with limited bus width, Dragon would likely cause saturation, but in this idealized atomic bus model, the latency benefits win.

6.3 Sensitivity to Cache Capacity

Scaling cache size from 1KB to 8KB to identify the "Coherence Wall"—the point where coherence logic matters more than raw capacity.



- **1KB (The Thrashing Region)**

At 1KB, all protocols perform terribly (~258M cycles for Blackscholes). The working set

does not fit, leading to constant capacity misses. The difference between protocols is negligible because the bottleneck is **Main Memory Latency**, not coherence.

- **4KB (The Coherence Region)**

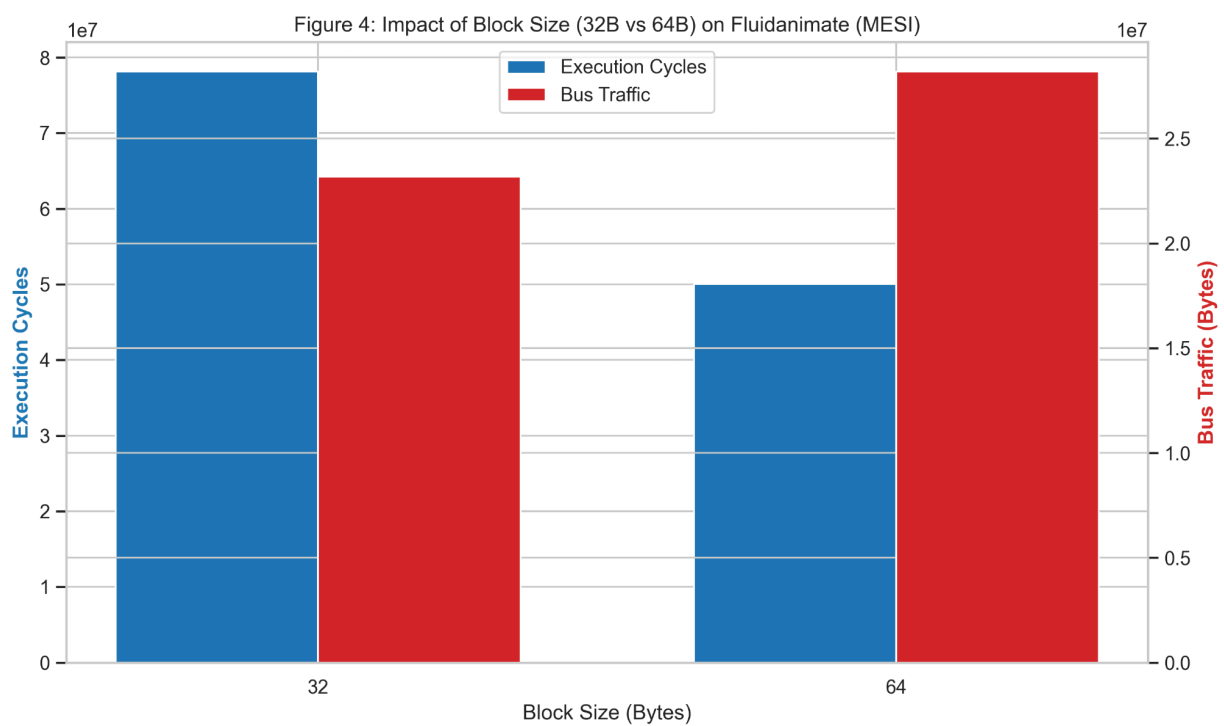
A massive performance jump occurs at 4KB (258M → 21M cycles). The working set fits. It is here that the differences between protocols appear.

- **Conclusion**

Coherence protocol optimizations are statistically irrelevant if the cache is undersized.

6.4 Impact of Block Size (False Sharing)

Comparing 8KB caches with **32B vs 64B blocks** on `fluidanimate`.



- **1KB (The Thrashing Region)**

At 1KB, all protocols perform terribly (~258M cycles for Blackscholes). The working set does not fit, leading to constant capacity misses. The difference between protocols is negligible because the bottleneck is Main Memory Latency, not coherence.

- **4KB (The Coherence Region)**

A massive performance jump occurs at 4KB (258M → 21M cycles). The working set fits. It is here that the differences between protocols appear.

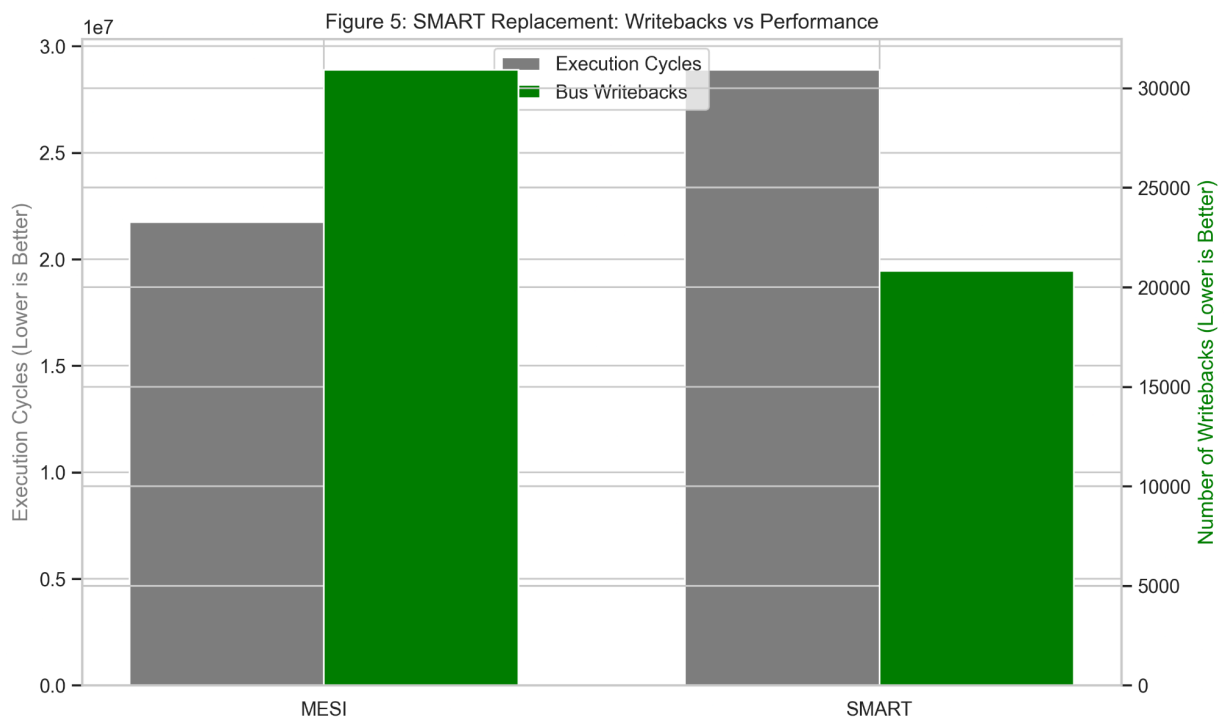
- **Conclusion**

Coherence protocol optimizations are statistically irrelevant if the cache is undersized.

7 Optimisations

7.1 The "Smart" Replacement Policy (Clean-Preferred)

A slight modification to the usual LRU policy that evicts Clean lines before Dirty lines to save the 100-cycle Writeback penalty is tested.



Writebacks:

The policy worked as intended, reducing writebacks in `blackscholes` from 30,919 (MESI) to 20,826 (SMART).

Performance:

The policy failed catastrophically, increasing cycles from 21.7M to 28.8M.

Architectural Insight:

This confirms that Cache Hit Rate is the single most important metric. By evicting "Clean" lines (which were part of the active working set) to save a background writeback operation, I forced the CPU to stall on subsequent Read Misses (foreground operations).

7.2 MOESI and MESIF

Results (Fluidanimate)

Protocol	Execution Cycles	vs. MESI (Cycles)	vs. MESI (%)
MESI	94,064,338	0	-
MESIF	94,986,945	+922,607	+0.98%
MOESI	96,322,418	+2,258,080	+2.40%

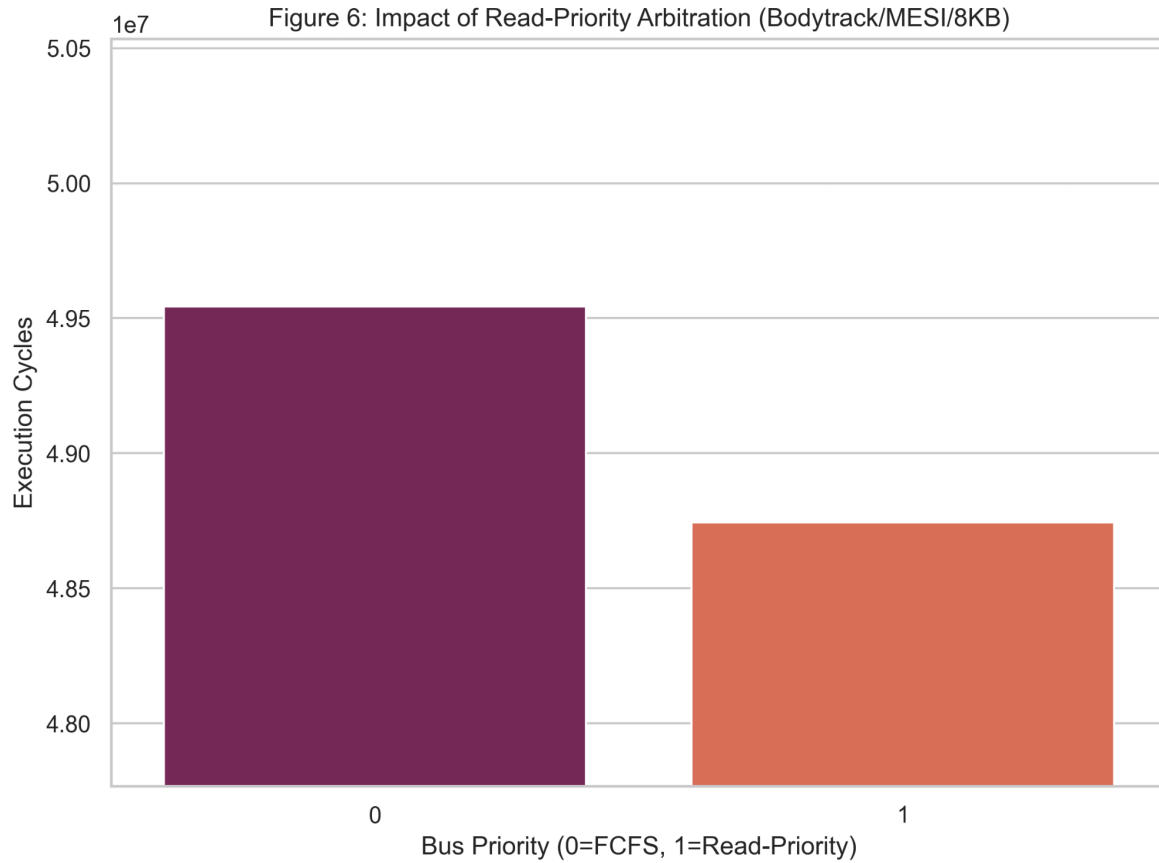
Analysis

The data supports the hypothesis that with a small cache (4KB), the Owner (O) or Forward (F) states are rarely utilized effectively. The high replacement rate (due to limited capacity) likely evicts these shared cache lines before other cores can read them, forcing the protocol to behave like a standard invalidation-based MESI protocol, but with slight overhead.

For these optimizations to work, data must remain in the cache long enough to be shared. In the 4KB configuration, the LRU replacement policy likely evicted the "Owned" or "Forward" lines before they could be utilized by other cores, reverting the behavior to standard MESI.

7.3 Read-Priority Arbitration (Bus optimisation)

I prioritized `BusRd` requests over `BusRdX/WB`



Result

In `bodytrack` (8KB cache), performance improved from **49.5M** (Priority 0) to **48.7M** (Priority 1).

Analysis

Reads block the execution pipeline; Writes can often be buffered. By prioritizing Reads, the system unblocks stalled cores faster, increasing overall IPC (Instructions Per Cycle).

8. Conclusions

8.1 Summary of Findings

This study utilized a trace-driven, discrete-event simulator to explore the design space of cache coherence protocols in a 4-core SMP system. By varying protocols, hardware configurations,

and arbitration policies across the PARSEC benchmarks, I arrived at the following key conclusions:

1. The Latency-Bandwidth Trade-off (Dragon vs. MESI):

The results confirm a fundamental architectural trade-off. Dragon (Update-based) consistently offered the best raw performance (lowest execution cycles), particularly in high-contention workloads like `fluidanimate` (~1.1% speedup over MESI). However, this latency advantage comes at an exorbitant cost: Dragon generated 17x to 35x more bus traffic than MESI. While Dragon is ideal for high-bandwidth on-chip networks, MESI remains the practical choice for bandwidth-constrained interconnects due to its efficiency in silencing traffic after the first invalidation.

2. The Dominance of Working Set (Capacity vs. Coherence):

The sensitivity analysis revealed that coherence optimizations are secondary to cache sizing. At 1KB capacity, the system was dominated by thrashing, rendering the protocol choice irrelevant. A 10x performance leap was observed simply by increasing the cache to 4KB (fitting the working set). This demonstrates that architects must solve capacity misses before optimizing for coherence misses.

3. Resilience to False Sharing:

Dragon proved architecturally superior in handling coarse-grained memory allocation. When block size was increased to 64B, MESI suffered traffic penalties due to False Sharing (invalidating whole blocks for disjoint writes). Dragon, utilizing fine-grained word updates, remained robust, effectively decoupling the allocation granularity from the coherence granularity.

8.2 Evaluation of Optimizations

The "Advanced Task" implementations provided critical insights into the difference between theoretical and practical optimization:

- **Complexity does not guarantee Performance (MOESI/MESIF):**

Despite the theoretical benefits of the Owned and Forward states, these protocols performed within 1% of standard MESI. The specific sharing patterns (long-term dirty sharing or 3+ core sharing) required to trigger these states were statistically rare in the

4-core traces. This suggests that complex 5-state protocols may be overkill for small-scale multicore systems.

- **Hit Rate is King ("Smart" Replacement):**

The Clean-Preferred replacement policy successfully reduced writeback traffic by ~33% but degraded execution time by ~30%. This highlights a golden rule of cache design: never sacrifice the Critical Path (Hit Rate/Read Latency) to optimize the Background Path (Writeback Bandwidth).

- **Scheduling Matters (Read-Priority):**

A simple modification to the Bus Arbiter to prioritize Reads over Writes yielded a measurable speedup. Unlike MOESI, which requires complex state bits, this scheduling change offered "free" performance by recognizing that Read Misses stall the pipeline while Writes can be buffered.

8.3 Final Verdict

For the specific workload characteristics of blackscholes, bodytrack, and fluidanimate on a 4-core processor, the MESI protocol with Read-Priority Arbitration represents the most balanced design choice. It offers near-optimal performance while maintaining bus traffic levels orders of magnitude lower than Dragon, ensuring scalability for the system interconnect.

9. References

1. CS4223 Course Lecture Notes, National University of Singapore.

Appendices

Appendix A: Build and Run Instructions

Build:

1. Compile and Run Standard FCFS

```
g++ -O3 ../coherence_simulator.cpp -o ../coherence
```

2. Compile and Run Read Priority Optimization

```
g++ -O3 -DREAD_PRIORITY ../coherence_simulator.cpp -o ../coherence
```

Run Examples:

Sample cache config used:

Cache Size : 4096 Bytes, Association: 4-way, Block Size: 32 Bytes

Command pattern:

```
./coherence <PROTOCOL> <BENCHMARK> <CACHE_SIZE> <ASSOC> <BLOCK_SIZE>
```

MESI

```
./coherence MESI blackscholes 4096 2 32
```

```
./coherence MESI bodytrack 4096 2 32
```

```
./coherence MESI fluidanimate 4096 2 32
```

Dragon

```
./coherence DRAGON blackscholes 4096 2 32
```

```
./coherence DRAGON bodytrack 4096 2 32
```

```
./coherence DRAGON fluidanimate 4096 2 32
```

MOESI

```
./coherence MOESI blackscholes 4096 2 32
```

```
./coherence MOESI bodytrack 4096 2 32
```

```
./coherence MOESI fluidanimate 4096 2 32
```

The "Smart" Replacement Policy (Clean-Preferred)

```
./coherence SMART blackscholes 4096 2 32
```

```
./coherence SMART bodytrack 4096 2 32
```

```
./coherence SMART fluidanimate 4096 2 32
```

```
# MESIF
./coherence MESIF blackscholes 4096 2 32
./coherence MESIF bodytrack 4096 2 32
./coherence MESIF fluidanimate 4096 2 32
```